

- PAULA MENCHÓN, *Partition contact algebras*.

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Boolean Contact Algebras (BCAs) are Boolean algebras expanded with a binary relation C . Under the intended interpretation, elements of the domain represent regions of space, and C denotes the contact relation between regions. BCAs were introduced as a point-free approach to topology and, over the last 25 years, have developed into a vibrant area of research at the intersection of logic and mathematics.

In this presentation, we focus on a particular class of BCAs arising from partitions of the unit [1]. Given such a partition, a contact relation can be defined as follows: two regions x and y are in contact if and only if they both overlap the same cell of the partition. This subclass of BCAs is referred to as *Partition Contact Algebras* (PCAs).

PCAs admit a natural interpretation. When the partition is finite, it can be viewed as a spatial resolution, with its elements playing the role of pixels. In this setting, the contact relation represents a rough or approximate notion of nearness between regions, relative to the coarseness of the resolution: the finer the partition (that is, the greater the number of pixels), the fewer regions are in contact with one another.

The talk will concentrate on the algebraic properties of Partition Contact Algebras. This is a joint work with Sergio Celani and Rafał Gruszczyński (Toruń, Poland).

[1] GRUSZCZYŃSKI, R. AND MENCHÓN, P., *From contact relations to modal operators, and back*, *Studia Logica*, vol. 111(2023), no. 5, pp. 717–748.